

BLACK-SCHOLES MODEL AND HEAT KERNEL WITH
APPLICATION IN FINANCE

BLACK-SCHOLES 模型和热核及其在金融中的应用

A THESIS

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Abstract

The diffusion equation, also known as the heat equation, is one of the most fundamental partial differential equations (PDEs) in mathematics and physics. It plays an important role in describing the heat diffusion (or temperature distribution) over time within a given region. Although the diffusion equation was originally used to describe heat conduction, it has found widespread applications in finance, biology, chemistry, and engineering, particularly in modeling dynamic systems and stochastic processes. This paper aims to systematically explore the derivation methods of the diffusion equation, including the application of mathematical tools such as Fourier transforms and Green's functions. Special emphasis is placed on its applications in financial engineering, especially its relationship with the Black-Scholes model and the Fokker-Planck equation. By integrating theoretical derivations with practical applications, this study highlights the significant role and future potential of the heat kernel in modern finance.

扩散方程，又称热方程，是数学和物理学中最基本的偏微分方程之一。它在描述给定区域内随时间的热扩散（或温度分布）方面起着重要作用。虽然扩散方程最初是用来描述热传导的，但它已经在金融、生物、化学和工程中得到了广泛的应用，特别是在动态系统和随机过程的建模中。本文旨在系统地探讨扩散方程的推导方法，包括傅里叶变换与 Green's 函数等数学工具的引入，并重点分析其在金融工程领域中的应用，尤其是与 Black-Scholes 模型及 Fokker-Planck 方程的关联。通过理论推导与实际应用相结合，揭示热核在现代金融中的重要作用与发展前景。

KEY WORDS: diffusion equation, heat equation, heat kernel, Fourier transforms, Black-Scholes

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Chapter 1

Introduction

The diffusion and heat equations are fundamental mathematical models that describe the distribution of physical quantities such as temperature, concentration, or probability density over time. These equations originated in physics, especially through Fourier's pioneering work on heat conduction, and have permeated many fields, including probability theory, engineering and financial mathematics. Their solution not only provides insights into physical phenomena, but also lays the foundation for the pricing of financial derivatives and the modeling of stochastic processes.

Despite their long history and well-developed theoretical framework, the heat equation and related tools such as Fourier transforms and the heat kernel continue to inspire new research, especially in interdisciplinary applications. One of the most remarkable applications is in financial modeling, where the Black-Scholes equation, a cornerstone of modern quantitative finance, is a transformed version of the heat equation. Understanding the mathematical underpinnings of this transformation can provide deeper insight into option pricing and risk management.

This thesis aims to present a coherent exploration of the heat equation from three complementary perspectives: physical, probabilistic, and analytical. It first derives the heat equation from the energy conservation of one-dimensional rods, the random walk theory and intuitive physical explanations. It then introduces Fourier series and transforms, including the heat kernel, which plays a crucial role in solving these equations. Finally, the study applies these concepts to finance, particularly the Black-Scholes and Fokker-Planck equations.

The remainder of this thesis is organized as follows: Chapter 2 reviews the historical and theoretical foundations of the heat equation and its analytical methods. Chapter 3 presents the physical and probabilistic derivations of the diffusion equation. Chapter 4 discusses the Fourier analysis and Green's functions. Chapter 5 explores applications in finance. Chapters 6 and 7 conclude with discussion and summary.

This study ultimately focuses on how the heat kernel and related mathematical methods are applied in the context of the Black-Scholes model, highlighting the deep interplay between mathematical theory and financial practice.

Chapter 2

Literature Review

The heat equation has been one of the most popular topics of study in physics and mathematics and since the early 1800s. The discovery of heat conduction by Fourier in 1810 is a significant development. Fourier's three-part paper *On the Propagation of Heat in Solid Bodies* (1822) is considered to be the basis of modern analysis. To solve the heat equation, Fourier used a concept of expanding functions in terms of trigonometric series, which is called Fourier series. It is quite obvious that Fourier's approach of breaking down the heat patterns into simple sine and cosine waves transformed the approach of solving partial differential equations (PDEs) [9]. This paper continues the work of the original Fourier and emphasizes applications in problems of financial engineering.

Research work on George Green in 1828 also marks another important point in the history of the diffusion equation. In his influential work, Green described Green's functions and their application in solving boundary value problems, which played a significant role in advanced-level studies of PDEs [10]. He also applied potential functions to the theories of electricity and magnetism and formulated, systematically using them to solve boundary value problems in physics. Although it had been introduced and used by Lagrange, Laplace, and Poisson, it was Green who gave a proper form to the theory of potential functions and made them useful in solving actual problems [10].

Another great mathematician of the 18th century was Jean le Rond d'Alembert, who suggested the d'Alembert equation for describing one-dimensional wave motion, which helped advance the knowledge of heat and diffusion equations [11]. He contributed many ideas to fluid mechanics, acoustics, and dynamic processes in physical systems. Jean-Marie Constant Duhamel later developed further methods for solving non-homogeneous partial differential equations with Duhamel's principle, which was later extended for the solution of non-homogeneous heat equations as well as other evolution equations with time-dependent sources [4].

Einstein also developed a probabilistic form of the heat equation as part of his research on Brownian motion [5]. In this way, Einstein established a probabilistic relation with the heat equation. This relation was put in mathematical context in the form of the Fokker-Planck equation formalizes this relationship, describing how probability distributions evolve in stochastic processes [8]. In finance, this probabilistic relation was used to describe and analyze the behavior of asset prices—an idea initially developed by Black, Scholes, and Merton, who applied stochastic

calculus to derive the analytical formula for pricing financial derivatives [3].

It has been used for several decades, and in recent years, people have gradually expanded these classical techniques to other applications, including using the heat kernel to solve boundary value problems and using Fourier transforms to solve multidimensional diffusion equations. As a fundamental solution to the heat equation, the heat kernel describes the spread of heat over time and plays a crucial role in finance and statistical physics. In financial engineering, for example, the Black-Scholes equation, which is applied to solve European option pricing problems, can be transformed into the heat equation and solved using the heat kernel, thus making the solution less costly [3]. Additionally, Leonhard Euler, a famous mathematician of the 18th century, contributed greatly to the evolution of heat kernel theory. Many mathematical concepts and tools developed by Euler have been essential in the understanding and application of the heat kernel, especially in solving complex boundary value problems [6].

Chapter 3

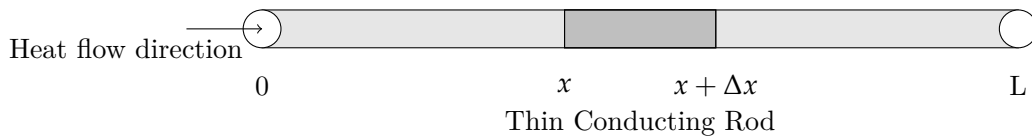
Diffusion Equation

3.1 Conservation Law of Energy for a One-Dimensional Rod

Before exploring the financial applications of the diffusion equation, we should first understand its physical origins and mathematical formulation. One classical method is to derive the equation from the conservation of energy applied to a one-dimensional conducting rod. This foundational derivation not only illustrates the physics behind the diffusion process but also serves as a basis for analogies in fields such as finance and stochastic modeling.

Now, We begin by considering a one-dimensional conducting rod with the following assumptions (see Farlow's book [7]) :

- The rod is made of a homogeneous conducting material.
- Heat flows only in the x -direction.
- The rod is thin, ensuring uniform temperature across any cross-section.



We now proceed to describe the energy change over an infinitesimal spatial interval $[x, x + \Delta x]$. The rate of energy change within this interval is given by the following equation:

$$\frac{d}{dt} \int_x^{x+\Delta x} c_p A U(x, t) dx = -A [F(b, t) - F(a, t)] \quad (3.1)$$

where:

- c_p : constant related to the specific heat of the material.
- A : cross-sectional area.
- $U(x, t)$: temperature distribution function.

- $F(x, t)$: heat flux.

Theorem 3.1 (Fourier's Law of Heat Conduction). *Let $U(x, t)$ be the temperature distribution at position x and time t . The relationship between the temperature gradient and the heat flux $F(x, t)$ at position x and time t is given by the following:*

$$F(x, t) = -k \frac{\partial U}{\partial x}(x, t)$$

where k is the thermal conductivity, a material-dependent constant.

By substituting Fourier's law into the net flux expression, we obtain:

$$c_p A \int_x^{x+\Delta x} U_t(x, t) dx = Ak [U_x(x + \Delta x, t) - U_x(x, t)] \quad (3.2)$$

Here, $U_t(x, t)$ denote the partial derivative of U with respect to time and $U_x(x, t)$ is the partial derivative of U with respect to x . To simplify the integral, we apply the Mean Value Theorem:

$$\int_a^b f(x) dx = f(\xi)(b - a)$$

Using this approximation for the integral in equation (3.2), we get:

$$c_p U_t(\xi, t) \Delta x = k \frac{U_x(x + \Delta x, t) - U_x(x, t)}{\Delta x} \quad (3.3)$$

Letting $\Delta x \rightarrow 0$, we obtain the standard form of the heat equation:

$$U_t(x, t) = \alpha^2 U_{xx}(x, t), \quad \alpha = \sqrt{\frac{k}{c_p}} \quad (3.4)$$

where α is the thermal diffusivity.

3.2 Random Walk

Next, we explore the relationship between the random walk model and the diffusion equation. We assume the following for the random walk:

- Time increment: Δt
- Space increment: Δx

The probability distribution at position X is symmetric, meaning the likelihood of moving right equals that of moving left

$$P_{right} = P_{left} = \frac{1}{2} \quad (3.5)$$

We knew at time $t + \Delta t$, the probability $P(X, t + \Delta t)$ at position X can be expressed as the average of the probabilities at positions $X + \Delta x$ and $X - \Delta x$ at time t :

$$P(X, t + \Delta t) = \frac{1}{2} P(X + \Delta x, t) + \frac{1}{2} P(X - \Delta x, t) \quad (3.6)$$

To calculate the difference in probability at position X between times $t + \Delta t$ and t , we have:

$$P(X, t + \Delta t) - P(X, t) = \frac{1}{2} P(X + \Delta x, t) + \frac{1}{2} P(X - \Delta x, t) - P(X, t)$$

Rearranging the terms:

$$P(X, t + \Delta t) - P(X, t) = \frac{1}{2} [P(X + \Delta x, t) - P(X, t)] + \frac{1}{2} [P(X - \Delta x, t) - P(X, t)] \quad (3.7)$$

To further simplify the difference terms, we also apply the Mean Value Theorem to both differences:

1. For $P(X + \Delta x, t) - P(X, t)$:

$$P(X + \Delta x, t) - P(X, t) = \Delta x \frac{\partial P}{\partial x}(X + \theta \Delta x, t), \quad 0 < \theta < 1 \quad (3.8)$$

2. For $P(X - \Delta x, t) - P(X, t)$:

$$P(X - \Delta x, t) - P(X, t) = -\Delta x \frac{\partial P}{\partial x}(X - \theta \Delta x, t), \quad 0 < \theta < 1 \quad (3.9)$$

Then, Substituting these expressions into the probability difference formula, we get:

$$P(X, t + \Delta t) - P(X, t) = \frac{1}{2} \left[\Delta x \frac{\partial P}{\partial x}(X + \theta \Delta x, t) \right] + \frac{1}{2} \left[-\Delta x \frac{\partial P}{\partial x}(X - \theta \Delta x, t) \right] \quad (3.10)$$

As $X + \theta \Delta x$ and $X - \theta \Delta x$ approach X , the difference can be approximated as:

$$P(X, t + \Delta t) - P(X, t) \approx \frac{\Delta x^2}{2} \frac{\partial^2 P}{\partial x^2}(X, t) \quad (3.11)$$

Dividing both sides by Δt and letting $\Delta t \rightarrow 0$, we finally obtain the diffusion equation:

$$\frac{\partial P}{\partial t} = \frac{\Delta x^2}{2 \Delta t} \frac{\partial^2 P}{\partial x^2} \quad (3.12)$$

This is the standard form of the diffusion equation corresponding to a random walk.

Dimensional Analysis

Let the dimensions of variables be:

- $[x] = L$ (length),
- $[t] = T$ (time),
- $[P] = Y$ (e.g., probability density or temperature).

Then:

$$\left[\frac{\partial P}{\partial t} \right] = \frac{[Y]}{[T]}, \quad \left[\frac{\partial^2 P}{\partial x^2} \right] = \frac{[Y]}{[L]^2}$$

Substituting into (3.12), we get:

$$\frac{[Y]}{[T]} = \left[\frac{\Delta x^2}{2 \Delta t} \right] \cdot \frac{[Y]}{[L]^2} \Rightarrow \left[\frac{\Delta x^2}{2 \Delta t} \right] = \frac{[L]^2}{[T]}$$

Therefore, the term $\frac{\Delta x^2}{2 \Delta t}$ has the dimension of a diffusion coefficient, often denoted by α :

$$\left[\frac{\Delta x^2}{2 \Delta t} \right] = [\alpha] = \text{length}^2 / \text{time}$$

This confirms that the right-hand side of the equation behaves as a physical diffusion operator, and $\alpha = \frac{\Delta x^2}{2 \Delta t}$ reflects the rate at which the quantity P spreads in space over time.

The heat equation, derived from the energy balance in a conducting rod, and the diffusion equation, arising from the random walk model, share a common mathematical structure. Both are described by a second-order partial differential equation governing the evolution of scalar quantities like temperature or probability over time and space. These connections highlight the broader applicability of diffusion processes in various physical systems.

3.3 Physical Intuition and Applications of Diffusion

Diffusion describes the process by which particles, energy, or other physical quantities spread from regions of higher concentration to regions of lower concentration over time. This natural phenomenon occurs because systems tend to evolve towards thermodynamic equilibrium, where differences in concentration, temperature, or other properties are minimized. Physically, diffusion can be understood as the random motion of particles (Brownian motion) leading to the net movement from crowded areas to less crowded ones.

Mathematically, the diffusion process is captured by the diffusion (or heat) equation, which describes how the density of the diffusing quantity changes over time and space. For instance, the diffusion of dye particles in water can be modeled by a partial differential equation that accounts for the random microscopic motion of particles and their cumulative macroscopic effect. In the limit of many small random movements, the Central Limit Theorem ensures that the resulting particle distribution approaches a Gaussian profile, a characteristic feature of diffusion.

The diffusion equation finds widespread applications across various scientific fields. In thermodynamics, it governs heat conduction in solids. In finance, it underpins models of asset price evolution, such as in the derivation of the Black-Scholes equation. In biology, it models the transport of nutrients and signaling molecules within cells. In environmental science, diffusion equations describe the spread of pollutants in air and water systems. These diverse applications highlight the fundamental role of diffusion processes in natural and engineered systems.

Chapter 4

Fourier Transform and Green's Function

The study of diffusion equations and their solutions often naturally leads to Fourier analysis. Fourier methods provide a powerful set of tools for solving partial differential equations by transforming complex differential operations into simpler algebraic forms. In particular, the Fourier series expansion allows functions to be expressed as infinite sums of sines and cosines, making it possible to solve problems with periodic boundary conditions.

However, in many physical and financial models, the domain of interest is not bounded, and functions are defined over the entire real line. In such cases, the Fourier transform emerges as a natural extension of the Fourier series, enabling analysis on unbounded domains.

In this chapter, we first present the derivation from Fourier series to Fourier transform, and then introduce the Green's function approach for solving partial differential equations, particularly focusing on its application to the diffusion equation.

4.1 Derivation from Fourier Series to Fourier Transform

The Fourier series provides a powerful method for representing a function defined on a finite interval, such as $[-L/2, L/2]$, as a sum of sine and cosine functions. This decomposition reveals the frequency content of a periodic signal and serves as a foundational tool in fields such as signal processing, vibration analysis, and image analysis.

However, many real-world signals are not periodic, or they may be defined over the entire real line rather than a finite interval. To analyze such non-periodic functions, we need a more general tool: the Fourier Transform.

The Fourier Transform emerges as a natural extension of the Fourier Series by considering the limit as the interval length $L \rightarrow \infty$. In this limit, the discrete set of frequencies becomes a continuous spectrum, and the summation in the Fourier series transitions into an integral. This process not only generalizes the Fourier series but also establishes a deep connection between time-domain and frequency-domain representations of functions.

In this section, we will use Euler's formulas to express the Fourier series in terms of complex exponentials. Then, by taking the limit as $L \rightarrow \infty$, we will derive the expression for the Fourier

Transform. The approach presented here is inspired by articles published in the Zhihu Column [1, 2].

Definition 4.1.1 (Fourier Series). *Let $f(x)$ be a function defined on the finite interval $[-L/2, L/2]$, and assume that $f(x)$ satisfies the Dirichlet conditions. Then $f(x)$ can be represented by a Fourier series of the form:*

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{2n\pi x}{L}\right) + b_n \sin\left(\frac{2n\pi x}{L}\right) \right], \quad (4.1)$$

where the Fourier coefficients a_n and b_n are given by:

$$a_n = \frac{2}{L} \int_{-L/2}^{L/2} f(x) \cos\left(\frac{2n\pi x}{L}\right) dx, \quad (4.2)$$

$$b_n = \frac{2}{L} \int_{-L/2}^{L/2} f(x) \sin\left(\frac{2n\pi x}{L}\right) dx, \quad (4.3)$$

for $n = 1, 2, 3, \dots$, and

$$a_0 = \frac{2}{L} \int_{-L/2}^{L/2} f(x) dx. \quad (4.4)$$

Theorem 4.1 (Euler's Formula). *For any real number θ , the trigonometric functions $\cos \theta$ and $\sin \theta$ can be expressed in terms of complex exponentials as follows:*

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad (4.5)$$

$$\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \quad (4.6)$$

These identities are direct consequences of Euler's formula:

$$e^{i\theta} = \cos \theta + i \sin \theta. \quad (4.7)$$

By using the Euler's Formula 4.1. The Fourier series 4.1 can be rewritten in exponential form:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(\frac{a_n}{2} + \frac{b_n}{2i} \right) e^{i\frac{2n\pi x}{L}} + \left(\frac{a_n}{2} - \frac{b_n}{2i} \right) e^{-i\frac{2n\pi x}{L}} \quad (4.8)$$

Next, how can we combine the two parts of the Fourier series with different summation limits into a unified form? By observation, we can see that the original expression actually contains exponential terms with positive frequencies ($n = 1, 2, 3, \dots$) and negative frequencies ($n = -1, -2, -3, \dots$), as well as a constant term ($n = 0$). The coefficients of these terms can all be expressed in a unified way using complex numbers. Then we introduce a complex coefficient C_n , we get:

$$f(x) = \sum_{n=-\infty}^{\infty} C_n e^{i\frac{2n\pi x}{L}}, \quad \text{where } C_n \in \mathbb{C} \quad (4.9)$$

We can find when $n = 0$, $C_0 = \frac{a_0}{2}$.

To handle the coefficient C_n , if you've ever derived Fourier series coefficients, then naturally you'll need to use the concept of orthogonality. In the context of Fourier series, you're probably familiar with the orthogonality of sine and cosine functions. Here, we smoothly transition to the orthogonality of complex exponential functions.

Definition 4.1.2 (Orthogonality of Complex Exponentials). *Let $n, m \in \mathbb{Z}$ and $L > 0$. The set of complex exponential functions $\{\exp(i\frac{2\pi nx}{L})\}_{n \in \mathbb{Z}}$ is orthogonal over the interval $[-\frac{L}{2}, \frac{L}{2}]$, in the sense that:*

$$\int_{-L/2}^{L/2} \exp\left(-i\frac{2\pi mx}{L}\right) \exp\left(i\frac{2\pi nx}{L}\right) dx = \int_{-L/2}^{L/2} \exp\left(-i\frac{2\pi x}{L}(m-n)\right) dx = L\delta_{mn} \quad (4.10)$$

where δ_{mn} is the Kronecker delta, defined by

$$\delta_{mn} = \begin{cases} 1, & \text{if } n = m \\ 0, & \text{if } n \neq m \end{cases} \quad (4.11)$$

This orthogonality property is a key step in deriving the Fourier coefficients. It ensures that different frequency components do not interfere with each other when integrated, allowing us to isolate and solve for each individual coefficient C_n .

Suppose we multiply both sides of the Fourier series expansion of $f(x)$ by $\exp(-i\frac{2\pi mx}{L})$ and integrate over the interval $[-\frac{L}{2}, \frac{L}{2}]$:

$$\int_{-L/2}^{L/2} f(x) \exp\left(-i\frac{2\pi mx}{L}\right) dx = \sum_{n=-\infty}^{\infty} C_n \int_{-L/2}^{L/2} \exp\left(-i\frac{2\pi x}{L}(m-n)\right) dx \quad (4.12)$$

Using Definition 4.1.2, we know:

$$\int_{-L/2}^{L/2} \exp\left(-i\frac{2\pi x}{L}(m-n)\right) dx = L\delta_{mn} \quad (4.13)$$

Therefore, only the term where $n = m$ survives in the sum, and we obtain:

$$LC_n = \int_{-L/2}^{L/2} f(x) \exp\left(-i\frac{2\pi nx}{L}\right) dx \quad (4.14)$$

So, the Fourier coefficient C_n is given by:

$$C_n = \frac{1}{L} \int_{-L/2}^{L/2} f(x) \exp\left(-i\frac{2\pi nx}{L}\right) dx \quad (4.15)$$

Let $\xi = \frac{2\pi n}{L}$, then $\Delta\xi = \frac{2\pi}{L}$. the Fourier coefficient is written as:

$$LC_\xi = \int_{-L/2}^{L/2} f(x) e^{-i\xi x} dx \quad (4.16)$$

The function $f(x)$ can then be expressed as

$$f(x) = \sum_{n=-\infty}^{\infty} C_\xi e^{i\xi x} = \sum_{n=-\infty}^{\infty} \Delta\xi \frac{L}{2\pi} C_\xi e^{i\xi x} \quad (4.17)$$

Here, we have used a useful trick:

$$\Delta\xi \cdot \frac{L}{2\pi} = 1 \quad (4.18)$$

When dealing with Fourier series, a function defined on a finite domain is treated as being periodically extended beyond its original domain. This imposes an artificial periodicity. To handle general, non-periodic functions, we take the limit

$$L \rightarrow \infty \quad (4.19)$$

In this limit, the Fourier coefficient becomes

$$\lim_{L \rightarrow \infty} LC_{\xi} = \int_{-\infty}^{\infty} f(x) e^{-i\xi x} dx = \hat{f}(\xi) \quad (4.20)$$

Hence, the function $f(x)$ is recovered via the inverse transformation:

$$f(x) = \lim_{L \rightarrow \infty} \sum_{n=-\infty}^{\infty} \frac{\Delta \xi}{2\pi} (LC_{\xi}) e^{i\xi x} = \int_{-\infty}^{\infty} \frac{d\xi}{2\pi} \hat{f}(\xi) e^{i\xi x} \quad (4.21)$$

Here, $\hat{f}(\xi)$ is referred to as the Fourier transform of the function $f(x)$, and

$$\int_{-\infty}^{\infty} \frac{d\xi}{2\pi} \hat{f}(\xi) e^{i\xi x} \quad (4.22)$$

is known as the inverse Fourier transform.

Remark. *The definition of the Fourier transform is not necessarily unique.* Depending on the context (e.g., mathematics, physics, or engineering), the normalization constants and signs in the exponential can vary. For instance, some definitions distribute the 2π factor symmetrically between the forward and inverse transforms, or place it entirely in one direction. What remains invariant, however, is the duality and invertibility property of the transform.

We now formally define the Fourier transform and its inverse as follows:

Definition 4.1.3 (Fourier Transform and Inverse Fourier Transform). *Let $f(x)$ be a function in a suitable function space (e.g., $L^1(\mathbb{R})$ or $L^2(\mathbb{R})$). The Fourier transform of f is defined by*

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-i\xi x} dx. \quad (4.23)$$

Conversely, $f(x)$ can be recovered from its Fourier transform via the inverse Fourier transform:

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\xi) e^{i\xi x} d\xi. \quad (4.24)$$

The Fourier transform provides a rigorous and powerful framework for analyzing functions in the frequency domain. It plays an important role in many fields, including physics, applied mathematics, and signal processing.

4.2 Gaussian Functions and Fourier Transform

We now turn to a classic and very elegant example in Fourier analysis: the Fourier transform of the Gaussian function. It is worth noting that the Gaussian function is one of the few functions whose transformation maintains the same functional form.

Definition 4.2.1 (Gaussian Function). *The Gaussian function, also known as the normal distribution, is defined by*

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad (4.25)$$

where:

- $\mu \in \mathbb{R}$ is the mean (center) of the distribution,
- $\sigma^2 > 0$ is the variance, determining the spread.

For analytical convenience, we consider the case where $\mu = 0$, leading to the simplified form:

$$f(x) = Ne^{-\alpha x^2}, \quad (4.26)$$

with parameters defined as:

$$\alpha = \frac{1}{2\sigma^2}, \quad N = \sqrt{\frac{\alpha}{\pi}}. \quad (4.27)$$

The normalization constant N ensures that:

$$\int_{-\infty}^{\infty} f(x) dx = 1. \quad (4.28)$$

Substituting the simplified Gaussian function 4.26 into the transform 4.23, we have:

$$\hat{f}(\xi) = N \int_{-\infty}^{\infty} \exp[-\alpha x^2 - i\xi x] dx. \quad (4.29)$$

To evaluate this integral, we complete the square in the exponent:

$$\alpha x^2 + i\xi x = \alpha \left(x + \frac{i\xi}{2\alpha} \right)^2 - \frac{\xi^2}{4\alpha}. \quad (4.30)$$

Thus, the integral becomes:

$$\hat{f}(\xi) = Ne^{-\frac{\xi^2}{4\alpha}} \int_{-\infty}^{\infty} \exp \left[-\alpha \left(x + \frac{i\xi}{2\alpha} \right)^2 \right] dx. \quad (4.31)$$

We now perform a change of variables to simplify the integral:

$$x' = x + \frac{i\xi}{2\alpha}, \quad dx' = dx. \quad (4.32)$$

The new integration path is shifted into the complex plane:

$$\int_{-\infty + \frac{i\xi}{2\alpha}}^{\infty + \frac{i\xi}{2\alpha}} e^{-\alpha x'^2} dx'. \quad (4.33)$$

Since the integrand $e^{-\alpha x'^2}$ is entire (analytic everywhere in the complex plane) and decays rapidly as $|x'| \rightarrow \infty$, Cauchy's theorem allows us to deform the contour back to the real axis without changing the value of the integral. Therefore:

$$\int_{-\infty + \frac{i\xi}{2\alpha}}^{\infty + \frac{i\xi}{2\alpha}} e^{-\alpha x'^2} dx' = \int_{-\infty}^{\infty} e^{-\alpha x'^2} dx' = \sqrt{\frac{\pi}{\alpha}}. \quad (4.34)$$

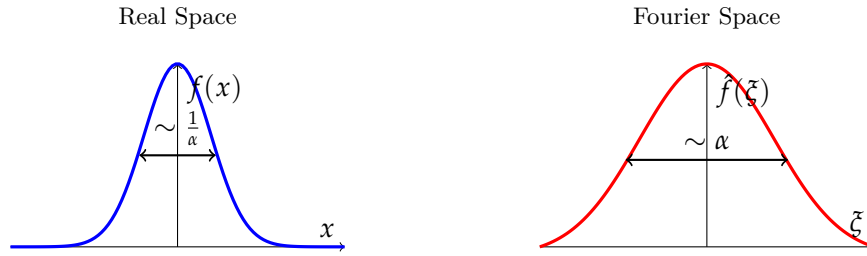
Hence, the final result is:

$$\hat{f}(\xi) = Ne^{-\frac{\xi^2}{4\alpha}} \cdot \sqrt{\frac{\pi}{\alpha}} = e^{-\frac{\xi^2}{4\alpha}}. \quad (4.35)$$

We conclude that the Fourier transform of a Gaussian is again a Gaussian, with the key feature that the “width” in frequency space is inversely related to the width in position space.

Duality and Visualization

The self-similarity of the Gaussian function under the Fourier transform illustrates a fundamental duality between the real and frequency domains. The narrower the function in one domain, the wider it becomes in the other. This is visualized below:



- If α is large (i.e., the original function is narrow), the Fourier transform is broad;
- If α is small (i.e., the original function is wide), the Fourier transform is narrow.

This reciprocal relationship is a direct manifestation of the “uncertainty principle”, which states that one cannot simultaneously localize a function in both time and frequency with arbitrary precision.

Limiting Cases

Two extreme limits further illustrate this duality:

- As $\alpha \rightarrow \infty$, the original function tends to a Dirac delta function $\delta(x)$, and its Fourier transform becomes a constant;
- As $\alpha \rightarrow 0$, the original function becomes very broad, and the Fourier transform tends to a Dirac delta function $\delta(\xi)$.

These limiting behaviors reflect the Fourier duality of a constant and a delta function.

Conclusion

The Gaussian function is invariant under the Fourier transform up to scaling. Its exceptional properties, including self-similarity, rapid decay, and analyticity, make it uniquely suited for applications in probability theory, quantum mechanics, signal processing, and beyond. Moreover, the Gaussian distribution plays a fundamental role in probability theory and appears widely in fields such as finance, where concepts like Markov chains and stochastic processes are closely tied to it.

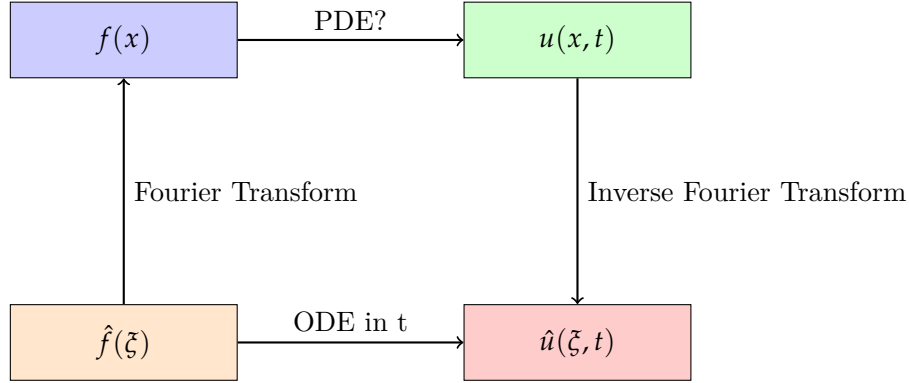
The inverse relationship between the Gaussian widths in real and frequency space exemplifies the uncertainty principle and underpins the foundational role of Gaussian in harmonic analysis. In many physical problems, such as electrostatics and heat diffusion, we are often tasked with solving partial differential equations (PDEs) that describe the distribution of some physical quantity, such as electric potential or temperature. In this chapter, we explore the important PDEs: the heat equation. We will discuss how the Fourier transform and Green's function techniques are used to solve these problems.

4.3 Fourier Transform Process

With the establishment of the theoretical basis of the Fourier transform, we now begin to discuss its application in solving partial differential equations. A classical and very important example

is the heat equation, whose Fourier transform leads to an explicit fundamental solution, often referred to as the Green's function or heat kernel. This result is of particular significance in both physics and quantitative finance.

First, we summarize the entire process through the following schematic diagram, which includes the basic steps for solving partial differential equations by applying the Fourier transform. This framework highlights how the original PDE in physical space is converted into an ordinary differential equation (ODE) in the frequency domain, which can then be solved and inverted to obtain the final solution.



The beauty of the Fourier transform is Δ becomes an algebraic “multiplier”. Below is a step-by-step explanation of its application to the homogeneous heat equation.

1. Problem Setup: The Heat Equation

In one spatial dimension, the heat equation is typically written as:

$$\begin{cases} (D.E) & u_t = ku_{xx}, \quad t > 0, \quad -\infty < x < \infty \\ (I.C.) & u(x, 0) = f(x), \quad -\infty < x < \infty \end{cases} \quad (4.36)$$

where $u(x, t)$ represents the temperature at position x and time t , $f(x)$ is the initial temperature distribution, and k is the thermal diffusivity constant. The equation $u_t = ku_{xx}$ models the diffusion of heat, where u_t is the time derivative of $u(x, t)$, and u_{xx} represents the second spatial derivative, describing the spatial diffusion of temperature.

2. Fourier Transform

To solve this partial differential equation (PDE), we apply the Fourier transform, a powerful mathematical tool that transforms the equation into an ordinary differential equation (ODE) in the frequency domain. This significantly simplifies the problem, as ODEs are generally easier to solve than PDEs.

Definition 4.3.1 (Fourier Transform). *The Fourier transform of a function $u(x, t)$ is defined as:*

$$\hat{u}(\xi, t) = \int_{-\infty}^{\infty} u(x, t) e^{-i\xi x} dx \quad (4.37)$$

The inverse Fourier transform is given by:

$$u(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{u}(\xi, t) e^{i\xi x} d\xi \quad (4.38)$$

Applying the Fourier transform to the heat equation $u_t = ku_{xx}$, we have:

The left-hand side, u_t , transforms as:

$$\hat{u}_t(\xi, t) = \frac{\partial}{\partial t} \hat{u}(\xi, t) \quad (4.39)$$

The right-hand side, u_{xx} , transforms as:

$$\widehat{u_{xx}}(\xi, t) = -(i\xi)^2 \hat{u}(\xi, t) = \xi^2 \hat{u}(\xi, t) \quad (4.40)$$

Thus, the equation becomes:

$$\begin{cases} (D.E) & \frac{\partial}{\partial t} \hat{u}(\xi, t) = -k\xi^2 \hat{u}(\xi, t), \quad t > 0, \quad -\infty < x < \infty \\ (I.C.) & \hat{u}(\xi, 0) = \hat{f}(\xi), \quad -\infty < x < \infty \end{cases} \quad (4.41)$$

This results is a simple ordinary differential equation (ODE) for each Fourier mode $\hat{u}(\xi, t)$. The solution to this ODE is:

$$\hat{u}(\xi, t) = \hat{f}(\xi) e^{-k\xi^2 t} \quad (4.42)$$

where $\hat{f}(\xi)$ is the Fourier transform of the initial condition $f(x)$.

The solution in the frequency domain represents the evolution of each Fourier mode. To find $u(x, t)$, we take the inverse Fourier transform of $\hat{u}(\xi, t)$:

$$u(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{u}(\xi, t) e^{i\xi x} d\xi \quad (4.43)$$

Substituting $\hat{u}(\xi, t) = \hat{f}(\xi) e^{-k\xi^2 t}$:

$$u(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\xi) e^{-k\xi^2 t} e^{i\xi x} d\xi \quad (4.44)$$

This integral expresses the solution as a sum of all Fourier modes, but it is not straightforward to compute directly.

3. Convolution Theorem

Next, applying the Convolution Theorem, we can express the solution (4.42) as follows:

$$\hat{u}(\xi, t) = \hat{f}(\xi) \hat{G}(\xi, t) \quad (4.45)$$

where

$$\hat{G}(\xi, t) = e^{-k\xi^2 t}. \quad (4.46)$$

The heat kernel $G(x, t)$, which is the inverse Fourier transform of $e^{-k\xi^2 t}$, is given by:

$$G(x, t) = G_t = \frac{1}{\sqrt{4\pi kt}} e^{-\frac{x^2}{4kt}} \quad (4.47)$$

Thus, the solution becomes:

$$u(x, t) = (f * G)(x, t) = \int_{-\infty}^{\infty} f(y) G(x - y, t) dy \quad (4.48)$$

This expression represents the solution to the heat equation as a weighted sum of the initial temperature distribution, where the weight function $G(x, t)$ governs the heat diffusion process over time. Overall, the final solution can be expressed as:

$$u(x, t) = \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} f(y) e^{-\frac{(x-y)^2}{4kt}} dy \quad (4.49)$$

This represents the temperature distribution at time t , and it shows how the initial temperature profile $f(x)$ spreads out over time due to heat diffusion. (see [12] for the source of idea)

4. Dimensional Analysis

$$\begin{aligned} \left[e^{-\frac{x^2}{4kt}} \right] = 1 &\implies \left[\frac{x^2}{4kt} \right] = 1 \implies [k] = L^2 T^{-1}. \\ [y] &= [L] \end{aligned}$$

The solution for the heat equation is given by:

$$\frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} f(y) e^{-\frac{(x-y)^2}{4kt}} dy.$$

This implies the dimensional consistency:

$$\frac{1}{[k]^{1/2} [t]^{1/2}} [f] [y] \approx [f] = [u].$$

The heat equation demonstrates the power of Fourier transforms in solving partial differential equations. By transforming the heat equation into the frequency domain, the problem is reduced to an ordinary differential equation, which is analytically solvable. The Fourier transform can be used to directly calculate the distribution of temperature over time, even under complex initial conditions. This technique provides an efficient method for solving the heat equation and has broad applications in physics, particularly in heat conduction and diffusion problems.

Chapter 5

Application of Heat Kernel in Finance

The heat kernel, as a fundamental solution to the heat equation, has extensive application in various fields, including mathematical finance. In this chapter, we explore how the heat kernel framework can be used to understand and solve key equations in quantitative finance.

Let's start with the Black-Scholes model, which provides a celebrated partial differential equation for option pricing. By transforming this equation into a heat equation, we can apply the heat kernel method to obtain analytical solutions and gain probabilistic insight into the dynamics of financial assets.

5.1 Black-Scholes Model

The Black-Scholes equation is given by:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0, \quad 0 < t < T \quad (5.1)$$

where S is the underlying asset price, V represents the option price, T is the expiration time, t is time, σ is volatility, and r is the risk-free interest rate. To simplify the Black-Scholes PDE, we make a change of variables using the natural logarithm of the asset price S . This substitution transforms the original equation into a more tractable form:

$$x = \log S \quad (5.2)$$

where x represents the logarithm of the price of the underlying asset. With this substitution, we need express the partial derivatives of V with respect to S and t in terms of those with respect to x and t . To do this, we first apply the chain rule to compute the required derivatives.

$$\frac{\partial V}{\partial S} = \frac{\partial V}{\partial x} \cdot \frac{\partial x}{\partial S} = \frac{\partial V}{\partial x} \cdot \frac{1}{S} \quad (5.3)$$

$$\frac{\partial^2 V}{\partial S^2} = \frac{1}{S^2} \left(\frac{\partial^2 V}{\partial x^2} - \frac{\partial V}{\partial x} \right) \quad (5.4)$$

Substituting back into the Black-Scholes equation, we obtain the following transformed equation:

$$\frac{\partial V}{\partial t} + \left(r - \frac{\sigma^2}{2}\right) \frac{\partial V}{\partial x} + \frac{\sigma^2}{2} \frac{\partial^2 V}{\partial x^2} = rV \quad (5.5)$$

Observing the structure of the Black-Scholes equation (5.1), we recognize it closely resembles a Cauchy-Euler equation. So, we can also apply the Cauchy-Euler equation to solve this equation.

Theorem 5.1 (Cauchy-Euler Equation). *Consider the Cauchy-Euler equation:*

$$aS^2 \frac{\partial^2 V}{\partial S^2} + bS \frac{\partial V}{\partial S} + cV = 0,$$

where a , b , and c are constants. The solution to this equation is of the form $V(S) = S^m$, where m satisfies the quadratic equation:

$$am^2 + (b - a)m + c = 0.$$

By using this theorem, we can get the same equation (5.5), if we isolate the first and third terms, we obtain:

$$\frac{\partial V}{\partial t} \approx \frac{\sigma^2}{2} \frac{\partial^2 V}{\partial x^2}. \quad (5.6)$$

This can be interpreted as:

$$\tau \equiv \frac{\sigma^2}{2}(T - t), \quad (5.7)$$

where τ represents the transformed time to maturity. This transformation scales the time variable by a factor related to the volatility σ , and changes the equation into a form more amenable to standard solution techniques. The transformed equation now becomes:

$$\frac{\partial V}{\partial \tau} = \left(\frac{2r}{\sigma^2} - 1\right) \frac{\partial V}{\partial x} - \frac{2r}{\sigma^2} V + \frac{\partial^2 V}{\partial x^2}. \quad (5.8)$$

This is a second-order linear PDE for V with respect to the transformed variables x and τ . The equation is now a standard form for which well-established solution methods exist. To solve the transformed Black-Scholes equation, we introduce an auxiliary function $U(x, \tau)$ by setting:

$$V(x, \tau) = e^{ax+b\tau} U(x, \tau), \quad (5.9)$$

where $e^{ax+b\tau}$ is the integrating factor. As noted by my advisor, Andrew Lin, the integrating factor is a differential shift, which simplifies the equation by modifying its coefficients, often eliminating or simplifying the drift and volatility terms. This transformation reduces the PDE to a simpler form that can be solved more easily. By inserting this expression into the transformed equation, we can compute the derivatives of $V(x, \tau)$:

$$\frac{\partial V}{\partial \tau} = e^{ax+b\tau} \left(bU + \frac{\partial U}{\partial \tau}\right),$$

$$\frac{\partial V}{\partial x} = e^{ax+b\tau} \left(aU + \frac{\partial U}{\partial x}\right),$$

$$\frac{\partial^2 V}{\partial x^2} = e^{ax+b\tau} \left(a^2 U + 2a \frac{\partial U}{\partial x} + \frac{\partial^2 U}{\partial x^2} \right).$$

By substituting these derivatives into the transformed equation, we derive a new equation for $U(x, \tau)$:

$$\left(a + b - 2ar - 2r \frac{\sigma^2}{2} - a^2 \right) U + \left(1 - 2a - 2r \frac{\sigma^2}{2} \right) \frac{\partial U}{\partial x} + \frac{\partial^2 U}{\partial x^2} = 0. \quad (5.10)$$

For the equation to hold, we require the following conditions on the constants a and b :

$$a + b - 2ar - 2r \frac{\sigma^2}{2} - a^2 = 0,$$

$$1 - 2a - 2r \frac{\sigma^2}{2} = 0.$$

Solving these equations for a and b , we find the values:

$$a = \frac{1}{2} - \frac{\sigma^2}{2r}, \quad b = - \left(\frac{1}{2} + \frac{\sigma^2}{2r} \right)^2. \quad (5.11)$$

With these values of a and b , the function $U(x, \tau)$ satisfies the standard heat equation:

$$\frac{\partial U}{\partial \tau} = \frac{\partial^2 U}{\partial x^2}, \quad (5.12)$$

with the initial condition:

$$U(x, 0) = e^{-ax} F(e^x),$$

where $F(s)$ represents the payoff function.

Now that we know $U(x, \tau)$ satisfies the heat equation, we can apply the solution to the heat equation (4.49) .

$$U(x, \tau) = u(x, t) = \frac{1}{\sqrt{4\pi\tau}} \int_{-\infty}^{\infty} f(y) e^{-\frac{(x-y)^2}{4\tau}} dy$$

where $f(y)$ represents the initial condition. Using the standard result for the heat equation, the solution for $V(x, \tau)$ is:

$$V(S, t) = e^{-r(T-t)} \frac{1}{\sqrt{2\pi\sigma^2(T-t)}} \int_0^{\infty} e^{-\frac{(\log(S/S^*) + (r - \frac{\sigma^2}{2})(T-t))^2}{2\sigma^2(T-t)}} F(S^*) dS^*. \quad (5.13)$$

This is the closed-form solution for the price of a European call or put option in the Black-Scholes model. The integral represents the expected payoff under the risk-neutral measure, and the expression includes the discount factor $e^{-r(T-t)}$ to account for the time value of money.

5.2 the Fokker-Planck Equation

Next, we consider the Fokker-Planck equation, which governs the evolution of the probability density $u(x, t)$ with respect to both space and time. The equation is given by:

$$\begin{cases} (D.E.) & u_t = ku_{xx} + \gamma(xu)_x, & -\infty < x < \infty, t \geq 0, \\ (I.C.) & u(x, 0) = f(x), & -\infty < x < \infty \end{cases} \quad (5.14)$$

where $u(x, t)$ represents the probability density, k is the diffusion coefficient, γ is a constant related to the drift term, x , and $f(x)$ specifies the initial distribution at $t = 0$. To solve the equation, we apply the Fourier transform 4.3.1 with respect to the spatial variable x , defined as:

$$\hat{u}(\xi, t) = \int_{-\infty}^{\infty} u(x, t) e^{-ix\xi} dx,$$

Applying this to both sides of the Fokker-Planck equation, we obtain:

$$\hat{u}_t = -k\xi^2 \hat{u} + i\xi\gamma \int_{-\infty}^{\infty} xu(x, t) e^{-ix\xi} dx. \quad (5.15)$$

The second term in the Fokker-Planck equation involves the Fourier transform of the derivative $(xu)_x$, which simplifies to a derivative of $\hat{u}(\xi, t)$ with respect to ξ . This leads to the following first-order partial differential equation (PDE) for $\hat{u}(\xi, t)$:

$$\hat{u}_t + \gamma\xi\hat{u}_\xi = -k\xi^2\hat{u}. \quad (5.16)$$

This is a first-order linear PDE, which can be solved using the method of characteristics:

$$\frac{dt}{1} = \frac{d\xi}{\gamma\xi} = -\frac{d\hat{u}}{k\xi^2\hat{u}} \quad (5.17)$$

Then we have two first integrals:

$$\begin{aligned} \frac{dt}{1} &= \frac{d\xi}{\gamma\xi} \\ \frac{d\xi}{\gamma\xi} &= -\frac{d\hat{u}}{k\xi^2\hat{u}} \end{aligned}$$

By applying this method, we obtain two important first integrals of the form:

$$\xi e^{-\gamma t} = c_1, \quad \hat{u} e^{k\xi^2/2\gamma} = c_2, \quad (5.18)$$

where c_1 and c_2 are constants determined by the initial conditions. These integrals suggest that the solution depends on characteristic variables, which can be utilized to express the general solution for $\hat{u}(\xi, t)$ using Lagrange's theory for first-order PDEs.

According to the Lagrange's theory, the general solution is given by

$$\Phi(c_1, c_2) = \Phi\left(\xi e^{-\gamma t}, \hat{u} e^{\frac{k\xi^2}{2\gamma}}\right),$$

or by the implicit function theorem, there exists a function F such that

$$\hat{u}(\xi, t) e^{\frac{k\xi^2}{2\gamma}} = F(\xi e^{-\gamma t}).$$

where F is a function determined by the initial condition. By applying the initial condition $\hat{u}(\xi, 0) = \hat{f}(\xi)$, we find:

$$\hat{f}(\xi)e^{k\xi^2/2\gamma} = F(\xi). \quad (5.19)$$

Substituting this expression into the general solution, we obtain:

$$\hat{u}(\xi, t) = \hat{f}(\xi e^{-\gamma t})e^{-ka_\gamma(t)\xi^2}, \quad (5.20)$$

where $a_\gamma(t)$ is defined as:

$$a_\gamma(t) = \frac{1 - e^{-2\gamma t}}{2\gamma}. \quad (5.21)$$

As $\gamma \rightarrow 0$, $a_\gamma(t) \rightarrow t$, and the solution for $\hat{u}(\xi, t)$ approaches the solution for the classical heat equation. To obtain the solution $u(x, t)$ in real space, we apply the inverse Fourier transform 4.3.1 to $\hat{u}(\xi, t)$. First, we calculate the inverse Fourier transform of the Gaussian kernel:

$$K(x, t) = \mathcal{F}^{-1} \left(e^{-ka_\gamma(t)\xi^2} \right) = \frac{1}{\sqrt{4\pi ka_\gamma(t)}} e^{-\frac{x^2}{4ka_\gamma(t)}} = G(x, a_\gamma(t)). \quad (5.22)$$

This function represents the Green's function for the diffusion equation, which models how an initial distribution spreads out over time. Additionally, we have the following:

$$\mathcal{F}^{-1} \left[\hat{f}(\xi e^{-\gamma t}) \right] = e^{\gamma t} f(xe^{\gamma t}), \quad (5.23)$$

which accounts for the drift term in the solution. Thus, the complete solution for $u(x, t)$ is expressed as the convolution of these two terms:

$$u(x, t) = K(x, t) * e^{\gamma t} f(xe^{\gamma t}) = \int_{-\infty}^{\infty} K(x - y, t) e^{\gamma t} f(ye^{\gamma t}) dy. \quad (5.24)$$

Although the Fokker-Planck equation is not invariant under translations in x , the kernel $K(x, t)$ depends only on the difference $x - y$, which simplifies the problem into a standard convolution form.

Finally, we examine the limiting behavior as $\gamma \rightarrow 0$. As γ approaches zero, we observe that $a_\gamma(t) \rightarrow t$, and consequently, the kernel $K(x, t)$ converges to the Green's function of the classical heat equation:

$$K(x, t) \rightarrow G(x, t) = \frac{1}{\sqrt{4\pi kt}} e^{-\frac{x^2}{4kt}}. \quad (5.25)$$

Thus, in the limit as $\gamma \rightarrow 0$, the solution $u(x, t)$ converges to the solution of the classical heat equation:

$$\lim_{\gamma \rightarrow 0} u(x, t) = \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} f(y) e^{-\frac{(x-y)^2}{4kt}} dy. \quad (5.26)$$

This is the classical result for diffusion in the absence of drift, where the solution corresponds to the Gaussian distribution typically seen in heat conduction problems.

5.3 Future Prospects of Heat Kernel Applications

The heat kernel plays a pivotal role in financial mathematics, particularly in the modeling of stochastic processes and the solution of partial differential equations such as the Black-Scholes equation and the Fokker-Planck equation. By providing explicit solutions that describe the probabilistic evolution of asset prices, the heat kernel bridges the gap between analytical methods and stochastic modeling.

In the future, the application of thermonuclear methods in the financial field is expected to expand further. For example, thermonuclear technology can be used to solve more complex models involving random fluctuations, such as the Heston model. Furthermore, they offer potential in pricing special options, modeling credit risks through structural models, and analyzing high-frequency financial data applicable to diffusion approximations.

The universality and deep connection between thermonuclear and probability and analytical frameworks make it a powerful tool for solving the increasingly wide range of problems in quantitative finance.

Chapter 6

Discussion

The evolution and application of heat equations was one of the most significant advancements in mathematics and physical sciences in the last century, influencing many disciplines including financial engineering. Based on the brief introduction in the previous section, the heat equation also known as the diffusion equation, is developed based on the principles of energy conservation and probabilistic models of random walks and is applicable for one-dimensional systems. Despite being based on relatively simple assumptions, the model effectively explains a variety of complex processes in physics and other sciences. It has been used in thermodynamics, stochastic processes, financial engineering, and many other areas over the years.

The relation between diffusion processes and the heat equation is one more example of how different branches of mathematics interact in order to analyze real objects. As I have mentioned before, random walks help explain the process of diffusion and relate it to the heat equation. This version is particularly useful for explaining the role of the random motion of heat and, consequently, of the prices in the financial market. Through Fourier transforms and series the heat equation can be solved in the form of a boundary value problem. Not only does it help me explain the heat transfer, it also serves as a basis in estimating and analyzing the finance market, especially in derivative pricing.

To fully appreciate the mathematical underpinnings of these methods, this paper explored the derivation of the Fourier transform from the Fourier series. This transition highlights the conceptual leap from analyzing periodic functions to working with general aperiodic functions, a fundamental step in the formulation of continuous models used in heat and diffusion equations. Moreover, the special role of Gaussian functions under the Fourier transform was discussed, where their self-similarity under transformation underscores why they naturally arise in both the heat kernel and probability theory. This symmetry not only facilitates analytical solutions to the heat equation but also reinforces the connection between diffusion, uncertainty, and probabilistic modeling in finance.

Fourier, Green, d'Alembert, and Duhamel provided important contributions to history and developed the basis of the theory on which we work. It was a major improvement when Fourier came up with the concepts of Fourier series and transforms since it was by use of these that one could solve the heat equation. Not only did he solve the heat equation but he also gave tools that form the basis of many of the techniques used in signal processing, engineering, and

even in financial mathematics. Similarly, Green provided a theory of Green's functions and potentials, which gave the necessary method of solving diverse partial differential equations, which is crucial for the understanding of various physical and logical processes and the boundary value problems in many fields. These contributions paved the way for the formulation of more refined mathematical models that have found their application in numerous applications such as electromagnetism, fluid dynamics among others. Other contributions which helped to enhance the understanding of the heat equation solutions including d'Alembert and Duhamel's contributed in comprehending the solutions of the heat equation with time-dependent sources and non-homogeneous equations. It should therefore be possible to demonstrate how the solutions of heat equation can be applied to a number of practical situations, thus underlining the usefulness of the technique throughout physics and engineering disciplines.

The first relationship between these mathematical methods and the stock market appears with the heat equation that Black, Scholes and Merton applied to solve a physical problem. What is noteworthy is that they converted the Black Scholes equation into heat equation and thus were able to use the knowledge from mathematical physics to solve financial problems. This transformation shows how mathematical concepts work and how ideas from one branch of mathematics can be applied in another branch, in this case, option pricing. The relation between the heat equation and other stochastic processes in finance through the Fokker–Planck equation enhances this discussion. Stochastic processes, such as Brownian motion, are commonly used in finance to model asset price dynamics over time. Conversely, the heat equation describes the variations of probabilities and the behavior of complicated financial systems. This probabilistic method has numerous usages in the area of finance which includes the pricing of derivatives, measuring risk and estimating volatility of asset returns.

In the last few years, some more complicated higher dimensional problems have been solved with the aid of solutions of heat equations especially in the higher dimensional case. The heat kernel is a fundamental solution to the heat equation and is extensively used in various fields, including statistical mechanics, quantum mechanics, and finance. In high-dimensional scenarios, it is particularly valuable for solving boundary value problems, such as option pricing in financial decision-making and portfolio optimization. Thus, application of the heat kernel decreases the difficulty of solving these problems and makes the modelling more effective and efficient in financial applications.

Also, the use of Fourier transforms to solve multi-dimensional diffusion equations has brought new ways of solving complex problems in the area of pricing of derivative financial instruments, portfolio management and risk assessment. These methods allow the analysts to solve the high dimensional PDEs in easier way and may lead to the development of more elaborate models for determining asset values, risk management and hedging.

Euler significantly advanced the theories of infinite series and variational calculus; these tools helped in the formulation of theories of heat kernel. Some of his contribution in the understanding of infinite series and variational problems has been crucial in the analysis of complex PDEs as well as in the explanation of heat transfer in systems with varying boundary conditions. These tools are still crucial for contemporary developments of heat equation theory especially in the context of financial applications and computational methods.

Although the heat equation has been used widely and has many applications, there are still some areas for future research. New approaches in the computational methods, especially those based on the machine learning and artificial intelligence, may offer new ways for the solution of high-dimensional PDEs. In the field of finance these techniques might result in new developments in the option pricing and risk management specifically regarding the models with stochastic volatility and jump diffusion processes. There are also other stochastic processes that can be studied and their relation with diffusion models and the heat equation can be further explored. For instance, using the agent-based approaches or network theory, one can gain new insights into the diffusion of information and evolution of market prices.

Thus, it is possible to state that the heat equation, which has a long history of development and has numerous applications in physics, remains one of the most important in mathematics today. The cross-disciplinary character of its applications as well as the contributions of the great mathematicians to the history of the idea has made it highly effective in application to real-life problems. With new methods and techniques being discovered and implemented in the future, it is safe to assume that the heat equation and all concepts related to it will continue to be useful in the advancement of both theoretical and applied mathematics. As new techniques and technologies emerge, the heat equation and its related concepts will undoubtedly continue to play a vital role in shaping the future of both theoretical and applied mathematics.

Chapter 7

Conclusion

In conclusion, the diffusion equation is one of the most important mathematical formulations which find its application in a large number of fields such as thermodynamics, finance, biology, and many others. This paper has also focused on the derivation of the diffusion equation from different approaches such as energy conservation law, stochastic processes, Fourier analysis and the derivation and application of heat kernel as well as its practical uses in financial engineering. This paper also emphasizes the connections among diverse mathematical concepts and their possibilities of application in different disciplines. The contributions of Fourier, Green, d'Alembert and Duhamel laid a good foundation for the development of the diffusion equation while Black, Scholes and Merton applied them in finance to come up with modern financial engineering tools. Euler's mathematical insight played a great role in the advancement of complicated mathematical tools and it provided the necessary background for the application of heat kernel. Through these mathematical tools it is possible to give a more realistic description of the changes occurring in the real world, be it in the financial markets where asset prices are evolved or in the heat and diffusion processes. The diffusion equation and the related concepts have been proven to be of immense value and prospects in practical applications in real-life applications.

Bibliography

- [1] AIXR's 麦子, *Knowledge warehouse: Mathematics - fourier transform (part 1)*, 2022. Zhihu Column, accessed on 2025-04-28.
- [2] ———, *Knowledge warehouse: Mathematics - fourier transform (part 2)*, 2022. Zhihu Column, accessed on 2025-04-28.
- [3] F. BLACK AND M. SCHOLES, *The pricing of options and corporate liabilities*, Journal of Political Economy, 81 (1973), pp. 637–654.
- [4] J.-M. C. DUHAMEL, *Mémoire sur la propagation de la chaleur dans les solides*, Comptes Rendus de l'Académie des Sciences, (1837).
- [5] A. EINSTEIN, *Über die von der molekularkinetischen theorie der wärme geforderte bewegung von in ruhenden flüssigkeiten suspendierten teilchen*, Annalen der Physik, 17 (1905), pp. 549–560.
- [6] L. EULER, *Methodus inveniendi lineas curvas maximi minimive proprietate gaudentes*, 1744.
- [7] S. J. FARLOW, *Partial Differential Equations for Scientists and Engineers*, Dover Publications, unabridged, corrected, and enlarged republication ed. ed., 1993.
- [8] A. D. FOKKER, *Die mittlere energie der molekeln in einem idealen gase*, Zeitschrift für Physik, 20 (1914), pp. 351–360.
- [9] J.-B. FOURIER, *Théorie analytique de la chaleur*, Firmin Didot, 1822.
- [10] G. GREEN, *An Essay on the Application of Mathematical Analysis to the Theories of Electricity and Magnetism*, 1828.
- [11] J. LE ROND D'ALEMBERT, *Recherches sur la propagation du son*, 1747.
- [12] USER81375, *Green' s function for the heat equation*. Math StackExchange, 2021. <https://math.stackexchange.com/q/2589538>.